12.5 Probability Models and Expected Value

A **probability model** assigns a probability to every outcome in a sample space; it's valid only if every probability is between 0 and 1 and they all sum to exactly 1. **Theoretical probability** comes from a formula or equally-likely-outcomes assumption; **experimental probability** comes from actual trial data ($\frac{\text{occurrences}}{\text{trials}}$) — the Law of Large Numbers says the experimental value tends to approach the theoretical one as trials increase. **Expected value**, $E = \sum(\text{outcome} \times \text{its probability})$, is the long-run average result and is central to deciding whether a game, raffle, or insurance decision favors you or the house — even though it may never equal any single actual outcome.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Probabilities 0.1, 0.6, 0.3. Valid?

2. Probabilities 0.5, 0.5, 0.2. Valid?

3. 50 trials, event occurs 18 times. Find the experimental probability.

4. Fair die. Find the theoretical $P(6)$.

5. A game pays \$8 with probability 0.25, else \$0. Find E.

6. Cost \$2; expected payout \$2. Fair?

7. Cost \$3; pays \$12 with probability 0.2. Find the expected net.

8. Model: 0.4, 0.35, x. Find x.

9. 200 flips, 108 heads. Find the experimental $P(\text{heads})$.

10. Outcomes \$1 and \$5 with probabilities 0.8 and 0.2. Find E.

11. Cost \$5; expected payout \$4. Fair?

12. As trials increase, experimental probability tends toward what?

13. In a valid probability model, what must all outcome probabilities sum to?

14. 500 tickets sold; 1 prize of \$250 and 10 prizes of \$10 each. Find the expected winnings per ticket.

15. A \$500 laptop has an 8% chance of needing a \$300 repair. An extended warranty costs \$35. Using expected value, determine whether the warranty is a good deal on average, and by how much.

✓ Answer Key

1. Yes
2. No (sum = 1.2)
3. $\frac{9}{25}$
4. $\frac{1}{6}$
5. \$2
6. Yes
7. −0.60 (a loss of \$0.60)
8. 0.25
9. 0.54
10. \$1.80
11. No — favors the house
12. The theoretical probability
13. 1
14. \$0.70
15. Not a good deal on average; skipping it saves \$11 in expectation